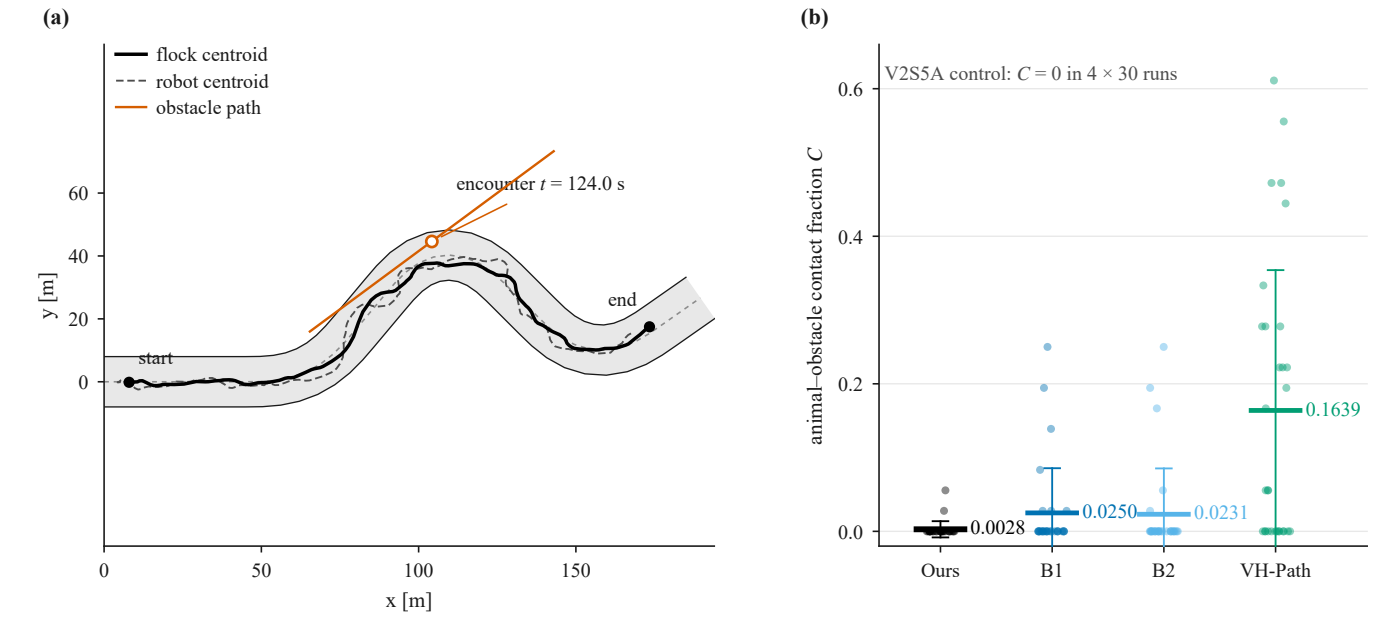

(a)
flock centroid
robot centroid
obstacle path
encounter t = 124.0 s
start
end
y [m]
x [m]
0
20
40
60
0
50
100
150
(b)
V2S5A control: C = 0 in 4 × 30 runs
animal–obstacle contact fraction C
0.0
0.2
0.4
0.6
0.0028
0.0250
0.0231
0.1639
Ours
B1
B2
VH-Path